



September 7, 2018

MT. DERM GmbH
% Susan D'Arcy
Director
Ismart Marketing Svcs
129 Green Lanes, Wylde Green
Birmingham, B735LT Gb

Re: K180778
Trade/Device Name: Exceed Microneedling Device
Regulation Number: 21 CFR 878.4430
Regulation Name: Microneedling device for aesthetic use
Regulatory Class: Class II
Product Code: QAI
Dated: July 26, 2018
Received: August 6, 2018

Dear Susan D'Arcy:

We have reviewed your Section 510(k) premarket notification of intent to market the device referenced above and have determined the device is substantially equivalent (for the indications for use stated in the enclosure) to legally marketed predicate devices marketed in interstate commerce prior to May 28, 1976, the enactment date of the Medical Device Amendments, or to devices that have been reclassified in accordance with the provisions of the Federal Food, Drug, and Cosmetic Act (Act) that do not require approval of a premarket approval application (PMA). You may, therefore, market the device, subject to the general controls provisions of the Act. Although this letter refers to your product as a device, please be aware that some cleared products may instead be combination products. The 510(k) Premarket Notification Database located at <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm> identifies combination product submissions. The general controls provisions of the Act include requirements for annual registration, listing of devices, good manufacturing practice, labeling, and prohibitions against misbranding and adulteration. Please note: CDRH does not evaluate information related to contract liability warranties. We remind you, however, that device labeling must be truthful and not misleading.

If your device is classified (see above) into either class II (Special Controls) or class III (PMA), it may be subject to additional controls. Existing major regulations affecting your device can be found in the Code of Federal Regulations, Title 21, Parts 800 to 898. In addition, FDA may publish further announcements concerning your device in the Federal Register.

Please be advised that FDA's issuance of a substantial equivalence determination does not mean that FDA has made a determination that your device complies with other requirements of the Act or any Federal statutes and regulations administered by other Federal agencies. You must comply with all the Act's requirements, including, but not limited to: registration and listing (21 CFR Part 807); labeling (21 CFR Part

801); medical device reporting (reporting of medical device-related adverse events) (21 CFR 803) for devices or postmarketing safety reporting (21 CFR 4, Subpart B) for combination products (see <https://www.fda.gov/CombinationProducts/GuidanceRegulatoryInformation/ucm597488.htm>); good manufacturing practice requirements as set forth in the quality systems (QS) regulation (21 CFR Part 820) for devices or current good manufacturing practices (21 CFR 4, Subpart A) for combination products; and, if applicable, the electronic product radiation control provisions (Sections 531-542 of the Act); 21 CFR 1000-1050.

Also, please note the regulation entitled, "Misbranding by reference to premarket notification" (21 CFR Part 807.97). For questions regarding the reporting of adverse events under the MDR regulation (21 CFR Part 803), please go to <http://www.fda.gov/MedicalDevices/Safety/ReportaProblem/default.htm>.

For comprehensive regulatory information about medical devices and radiation-emitting products, including information about labeling regulations, please see Device Advice (<https://www.fda.gov/MedicalDevices/DeviceRegulationandGuidance/>) and CDRH Learn (<http://www.fda.gov/Training/CDRHLearn>). Additionally, you may contact the Division of Industry and Consumer Education (DICE) to ask a question about a specific regulatory topic. See the DICE website (<http://www.fda.gov/DICE>) for more information or contact DICE by email (DICE@fda.hhs.gov) or phone (1-800-638-2041 or 301-796-7100).

Sincerely,

David Krause -S

for Binita S. Ashar, M.D., M.B.A., F.A.C.S.
Director
Division of Surgical Devices
Office of Device Evaluation
Center for Devices and Radiological Health

Enclosure

Indications for Use

510(k) Number (if known)

K180778

Device Name

Exceed Microneedling device

Indications for Use (Describe)

The Exceed is a microneedling device and accessories is intended for the treatment of wrinkles in Fitzpatrick skintypes I, II and/or III in the following facial areas: glabellar frown lines, periorbital lines and cheek folds in adults aged 22 years or older.

Type of Use (Select one or both, as applicable)

☒ Prescription Use (Part 21 CFR 801 Subpart D)

☐ Over-The-Counter Use (21 CFR 801 Subpart C)

CONTINUE ON A SEPARATE PAGE IF NEEDED.

This section applies only to requirements of the Paperwork Reduction Act of 1995.

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510k Summary

This 510(k) Summary is submitted in accordance with 21 CFR Part 807, Section 807.92(c).

Date Prepared: March 21st, 2018

Date Amended: September 6th 2018

Submitter's Name: MT.DERM GmbH

Submitter's Address: Gustav-Krone-Str. 3, 14167 Berlin, Germany

Submitter's telephone number: +49 30 845 885 171

Contact person: Andreas Pachten, DIRECTOR QM and Regulatory Affairs

Device Trade Name: Exceed Microneedling device

1. Device Classification Information:

Regulation Number	Device Classification name	Device Class	Product Code	Generic description	Classification Panel	Type
21 CFR 878.4430	Microneedling device for aesthetic use	Class 2	QAI	A microneedling device for aesthetic use is a device using one or more needles to mechanically puncture and injure skin tissue for aesthetic use. This classification does not include devices intended for transdermal delivery of topical products such as cosmetics, drugs, or biologics.	General & Plastic Surgery	Traditional 510 (k)

2. Device Description

The Exceed micro needling device is intended to create many, very tiny, microscopic punctures in the epidermal and dermal layers of the skin using sterile stainless-steel needles. The Exceed micro needling device consists of 7 component parts; Control Unit, Sterile single use Safety needle cartridge, Handpiece, Handpiece holder, Handpiece cover, Footswitch, Power supply

The control unit switches the device on and off and contains the power source (5.5-12 V, DC (150mA)). The control unit adjusts the frequency of the needle stroke from 100-150Hz using a digital display and a keypad. The control unit receives power via a coaxial connector from a standard 100-240V, 50-60Hz, 1.2A wall socket transformer with an output of 15V. The handpiece contains a motor that moves the needles and a needle protrusion dial that allows the user to control the depth of the needle protrusion. The handpiece contains a scale that allows for the needle protrusion depth to be adjusted by the operator. The scale has a tolerance of ± 0.15 mm. The standard safety needle cartridge is a 6-stainless steel micro needle plate of 1.5 mm length. The safety needle cartridge is screwed into the handpiece. The safety needle cartridge is sterile and for single use only. This standard

safety needle cartridge is used for microneedling only. The length of needles is 1.5 mm. In combination with the handpiece needle protrusion can be adjusted between 0.0 mm (min) and 1.5 mm (max).

3. Indications/Intended Use

The Exceed is a microneedling device and accessories is intended for the treatment of wrinkles in Fitzpatrick skin types I, II and/or III in the following facial areas: glabellar frown lines, periorbital lines and cheek folds in adults aged 22 years or older.

Caution: Federal law restricts this device to sale by or on the order of a physician.

4. Predicate Device

DEN160029 SkinPen Precision System

SkinPen Precision System DEN160029 is a microneedling device and accessories intended to be used as a treatment to improve the appearance of facial acne scars in adults aged 22 years or older.

The Exceed Micro needling device is predicated against the SkinPen Precision System because both devices are microneedling devices containing one or more needles to mechanically puncture and injure the skin tissue for aesthetic use.

5. Technological Characteristics

Property	Exceed Microneedling device	DEN160029 SkinPen Precision System	Significant differences
Device Manufacturer	MT.DERM GmbH Gustav-Krone-Str. 3, 14167 Berlin, Germany	Bellus Medical, LLC, 4505 Excel Parkway, Suite 100, Addison, TX 75001	Not applicable
Device Trade Name	Exceed Microneedling device	SkinPen Precision System	Not applicable
510(K) Number	K180778	DEN160029	Not applicable
Device Classification name	Microneedling device for aesthetic use	Microneedling device for aesthetic use	Identical
Device Product Code	QAI	QAI	Identical
Device Classification	Class II	Class II	Identical
Regulation number	21 CFR 878.4430	21 CFR 878.4430	Identical
Use	Prescriptive	Prescriptive	Identical
Intended Location of Use	Face	Face	Identical

Property	Exceed Microneedling device	DEN160029 SkinPen Precision System	Significant differences
Intended use and Indications	The Exceed is a microneedling device and accessories is intended for the treatment of wrinkles in Fitzpatrick skin types I, II and/or III in the following facial areas: glabellar frown lines, periorbital lines and cheek folds in adults aged 22 years or older.	SkinPen® Precision System is a microneedling device and accessories intended to be used as a treatment to improve the appearance of facial acne scars in adults aged 22 years or older.	Dissimilar – Clinical performance data provided demonstrates that the Exceed Microneedling device is safe and effective in the treatment of facial wrinkles in Fitzpatrick skin types I, II and/or III and raises no new issues in terms of safety and effectiveness.
Geometry	6 needles, squared arrangement	14 needles, radial arrangement	Dissimilar. Clinical and non-clinical performance testing demonstrates that the Exceed microneedling device does not pose any undue or additional risks and is effective.
Needle protrusion setting	0 - 1.9 mm	0 - 2.5 mm	Dissimilar. Clinical and non-clinical performance testing demonstrates that the Exceed microneedling device does not pose any undue or additional risks and is effective.
Needle protrusion setting	0 - 1.9 mm	0 - <2.7 mm ¹	Dissimilar. Clinical and non-clinical performance testing demonstrates that the Exceed Microneedling device does not pose any undue or additional risks and is effective.
Max Needle length used in clinical study	1.5mm	1.5mm	Substantially equivalent
Maximum needle penetration (maximum needle length)	1.5mm	2.5 mm	Dissimilar. Clinical and non-clinical performance testing demonstrates that the Exceed Microneedling device does not pose any undue or additional risks and is effective.

Property	Exceed Microneedling device	DEN160029 SkinPen Precision System	Significant differences
Frequency	100-150 Hz ($\pm 10\%$)	6300RPM to 7700 RPM ¹ (105 – 128.3 Hz)	Dissimilar. Clinical and non-clinical performance testing demonstrates that the Exceed Microneedling device does not pose any undue or additional risks and is effective.
Treatment protocol	4 treatments spaced 4 weeks apart	3 treatments spaced 4 weeks apart	Dissimilar – Clinical performance data provided demonstrates that the Exceed Microneedling device is safe and effective in the treatment of facial wrinkles, and raises no new issues in terms of safety and effectiveness.

¹ Data obtained from marketing material

6. Substantial Equivalency and Comparison of Technological Similarities & Differences

6.1. Key Similarities.

- i. The device classification (generic description) and basic technologies are equivalent in that both devices are micro needling devices containing >1 needle that mechanically punctures and injures the skin for aesthetic use.
- ii. Both devices are intended for prescriptive use.
- iii. The depth of penetration of the needles in clinical studies were equivalent – 1.5mm

6.2. Differences.

- iv. Although the devices share the basic generic description and technologies they do differ in several areas.
 - a. Intended use and indication
 - b. Geometry and needle count
 - c. Needle protrusion setting
 - d. Maximum needle length
 - e. Maximum needle penetration
 - f. Frequency
 - g. Treatment protocol

These differences have been addressed by the manufacturer through the applicable safety standards (General controls and mitigation measures) and through clinical and non-clinical performance testing (Special controls).

7. Non-clinical performance testing

To demonstrate safety and effectiveness and support substantial equivalence the Exceed Microneedling device has undergone non-clinical performance testing in line with recognized standards in terms of general requirements, biocompatibility, electrical safety and software.

In combination with the general controls of the FD&C Act, the microneedling device for aesthetic use is subject to the following special controls:

1. The technical specifications and needle characteristics are identified, including needle length, geometry, maximum penetration depth, and puncture rate.
2. Non-clinical performance data demonstrates that the device performs as intended under anticipated conditions of use.
The following performance characteristics are tested:
 - i. Accuracy of needle penetration depth and puncture rate in pig skin with high-speed camera measurement;
 - ii. Safety features built into the device to protect against cross-contamination, including fluid ingress protection due to a safety membrane; and
 - iii. Identification of the maximum safe needle penetration depth for the device in pig skin with high-speed camera measurement.
3. Performance data demonstrates the sterility of the patient-contacting components of the device according to ISO 11135-1:2007, ISO/TS 11135-2:2008 and ISO 11737-1:2006.
4. Performance data supports the shelf life of the device by demonstrating continued sterility, package integrity, and device functionality over the intended shelf life according to ISO 11607-1:2006+A1:2014 and ISO 11607-2:2006+A1:2014.
5. Performance data demonstrates the electrical safety and electromagnetic compatibility (EMC) of all electrical components of the device according to ISO 60601-1:2005+A1:2012 and ISO 60601-1-2:2014 (ed 4.0).
6. Software verification, validation, and hazard analysis were performed for all software components of the device according to ISO 62304:2006 (ed 1.0).
7. The patient-contacting components of the device were demonstrated to be biocompatible including evaluation of cytotoxicity, irritation, and sensitization, acute systemic toxicity and material-mediated pyrogenicity per the FDA guidance document "Use of International Standard ISO 10993-1, 'Biological evaluation of medical devices - Part 1: Evaluation and testing within a risk management process.
8. A cleaning validation was performed for reusable components of the device. (Reprocessing Medical Devices in Health Care Settings: Validation Methods and Labeling.)
9. Labeling included the following:
 - i. Information on how to operate the device and its components and the typical course of treatment;
 - ii. A summary of the device technical parameters, including needle length, needle geometry, maximum penetration depth, and puncture rate;
 - iii. Validated methods and instructions for reprocessing of any reusable components;
 - iv. Disposal instructions; and
 - v. Shelf life.
10. Patient labeling included:
 - i. Information on how the device operates and the typical course of treatment;
 - ii. The probable risks and benefits associated with use of the device; and
 - iii. Post-operative care instructions.

8. Clinical performance testing

A clinical study was conducted to support the safety and effectiveness of the Exceed Micro needling device for the treatment of facial wrinkles.

The objective of the study was to assess the effectiveness of the Exceed micro needling device in reducing the signs of skin aging as measured by the improvement in wrinkles, after 4 standardized treatment sessions and to assess the safety of the device as measured by the number of adverse events.

The study was conducted at a single center. Subjects were treated 4 times, 30 days apart (Day 0, 30, 60 and 90) with final assessment 60 days after the last treatment at 150 days.

Treatments were conducted by medically trained healthcare professionals.

The face was cleansed and a numbing cream containing a local anesthetic was applied to the skin for 30 minutes. After 30 minutes the numbing cream was removed from the patient's face and a small quantity of hydrogel was applied prior to the treatment to protect the skin against abrasion and friction during the procedure. The operator was instructed to start at the lowest needle depth setting and gradually increase the depth until pin point bleeding was observed with a maximum needle depth of 1.5 mm. Following the treatment, the skin was cleansed with warm water and sterile gauze. A physical sunblock ($SPF \geq 40$) was applied to the skin.

Demographics:

Forty-eight subjects were recruited for the study and 48 subjects completed the study. The average age of the subjects was 51.1 years (age range 39-67). Predominantly the population was female (1/48 (2%) subjects were male). The study included subjects with Fitzpatrick skin types I to III and V. None of the subjects were smokers (table i). No subjects with Fitzpatrick skin type IV or VI were included in the study.

No subject dropped out during the treatment phase of the study or during the follow-up. All subjects who were enrolled into the study and completed all treatment visits and all assessment visits were included in the assessment and analysis.

Table i – Demographic characteristics

Analysis	Age		M:F	Smoke	Fitzpatrick Skin type			
	Mean	Range			I	II	III	V
N = 48	55.1	39 - 67	1:47	0	3	36	8	1

Measurement of effectiveness

At each visit digital photographs (VISIA) were taken of the subject's face and the principal investigator assessed facial photodamage (including wrinkles) using a validated scale (Lemperle (2001)*. At the end of the study the digital images of the subject's face were randomized and analyzed independently by 3 physicians using the same grading scale (Lemperle (2001). The Lemperle grading scale divides the face into 9 regions and scores each facial area for the signs of aging using a photonic scale (0-5) referring to reference pictures where 0 indicates no wrinkles at rest and 5 being deep wrinkles at rest.

*Lemperle G, Holmes RE, Lemperle SS. A Classification of Facial Wrinkles. Plastic and reconstructive surgery. 2001 Nov 1;108(6):1735-50

Measurement of safety

Immediately after each micro needling procedure the principal investigator was asked to grade the amount of visible erythema in the treatment area. Grading was carried out using a 5-point grading scale;

- None: No erythema or redness. Skin is normal color.
- Minor: very faint erythema or redness. Skin has a slight redness to it.
- Mild: blotchy, visible redness that does not cover the entire face. Skin has a noticeable inconsistent redness to it.
- Moderate: dull red color to the skin. Skin has a very definite redness to it.
- Severe: Bright or dark red color to the skin. Skin is severely red.

Subjects also graded their erythema immediately after each micro needling procedure and in addition graded pain and discomfort experienced during the treatment procedure. A descriptive grading scale was used for subjects to evaluate their erythema with the addition of photography depicting grades of erythema in different phototypes.

This photonic scale was also used for subjects to record their erythema at home.

Pain and discomfort were recorded using an 11-point visual analogue scale (0-10) where 0 was equivalent to "no pain" or "no discomfort" to 10, "most intense pain ever" and "most discomfort ever". This grading structure was also used to record changes to pain and discomfort from the evening of the treatment to Day 8 (7 days after the treatment). In addition, skin peeling was assessed by the patient from day 3 to day 8 using a visual analogue scale where 0 represented "no skin peeling or flakiness" to 10 "heavy skin peeling over the whole face." During all times any potential adverse events were recorded by the research staff.

Results - Safety – Physician reported outcomes

In 13 out of 48 subjects (27%) adverse events were reported during the study (table ii).

Nine (9) adverse events were related to the treatment. Eight (8) were herpes simplex labialis (HSL) outbreaks and 1 adverse event was dry skin. Of the 8 HSL outbreaks, 5 occurred in subjects with Fitzpatrick skin type 2, 2 occurred in skin type 3 and 1 occurred in skin type V. One subject (skin type 2) reported extremely dry skin which was self-limiting and not reported again by the subject.

Note: Thirteen (13) out of 48 subjects (27%) reported a predisposition to HSL at the beginning of the study. Of this predisposed population, an HSL outbreak was observed in 8 subjects (62%). Subjects were treated for their HSL using conventional antiviral therapy and thereafter were given prophylactic antiviral therapy prior to the micro needling procedure. There were no further reported outbreaks.

Table ii. Adverse event - summary of observations and rate

Event	Severity of adverse event						Total
	Mild		Moderate		Severe		
	Related to the device (R)	Not-related to the device (NR)	Related	Non-related	Related	Non-related	
Herpes Labialis n=8	8 (100%)	0	0	0	0	0	8
Dry skin n=1	1 (100%)	0	0	0	0	0	1
Cold/Flu n=2	0	2 (100%)	0	0	0	0	2
Surgery n=2	0	2 (100%)	0	0	0	0	2
Total	9	4	0	0	0	0	13

Following treatment, 48/48 (100%) of subjects experienced erythema. Physician grading of erythema immediately after the treatment was recorded as minor in 70% (34/48) of subjects. The remainder of subjects were graded as moderate by the physician.

Results - Safety – Patient reported outcomes

Subjects were also asked to grade their erythema immediately at the end of the treatment. Nineteen out of 48 (40%) subjects graded their skin redness as moderate and 29/48 (60%) subjects as severe (table iii).

Table iii. Erythema assessment by physician and subject immediately after treatment

Grading of erythema immediately after treatment	Grading		
	Minor	Moderate	Severe
Physician	34/48 (70%)	14/48 (30%)	0
Subject	0	19/48 (40%)	29/48 (60%)

Subjects indicated that although the treatment appeared to be moderately painful it was not particularly uncomfortable (table iv).

Table iv. Pain and discomfort experienced during the treatment

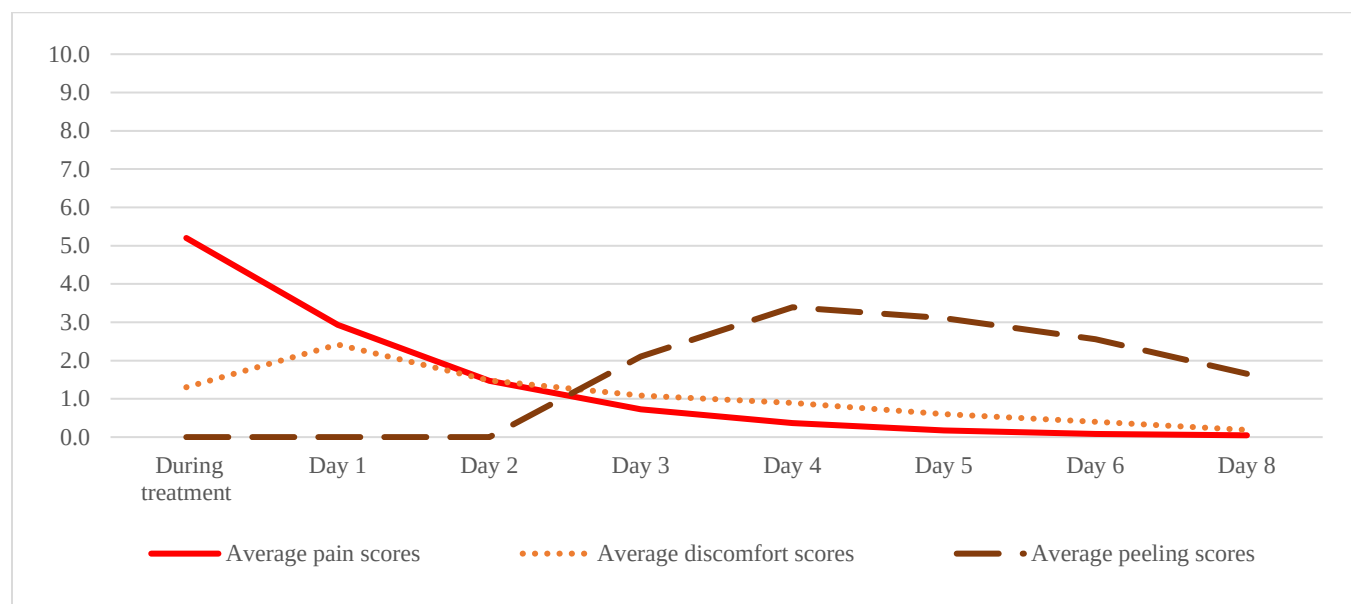
Subject assessment of pain and discomfort during the treatment (average over 4 treatments)	Pain score ¹ (range)
Pain during treatment	5.2 (3.8 to 6.1)
Discomfort during treatment	1.3 (<1 – 2.1)

¹. based on a scale 0-10 0 = No pain or discomfort. 10 = Worst pain or discomfort ever

Erythema, pain, discomfort and skin peeling were recorded up to 8 days after treatment.

- Erythema was experienced by 48 out of 48 subjects (100%) immediately after the treatment lasting 8 days. Eight days after the treatment 39/48 (81%) subjects reported no erythema, 9/48 (19%) subjects reported minor erythema and 1/48 (2%) subjects mild erythema.
- Pain was experienced by 48 out of 48 subjects (100%) to some degree during the treatment lasting 8 days. By day 3, 37/48 (77%) of subjects still reported pain. By day 8, 2/48 (4%) still reported some pain.
- Discomfort was experienced by 48 out of 48 subjects (100%) to some degree during the treatment lasting 8 days. Eight days after the treatment, discomfort was still reported by 9/48 (20%).
- Skin peeling was experienced by 48 out of 48 subjects (100%) from day 3 lasting 8 days. The amount of skin peeling reached a maximum of 3.4 day 4 (on a scale of 0-10 where 0 = no peeling and 10= heavy skin peeling over the whole face). Skin peeling was still reported in 31/48 (65%) subjects by the evening of day 8. figure i).

Figure i. Subject assessment of pain, discomfort and peeling immediately after the treatment to day 8



Results – Effectiveness – Physician reported outcomes

A blinded assessment by three blinded evaluators were performed. In accordance with the original clinical study the physicians used the grading structures of Lemperle (2001) for wrinkles. Lemperle's grading scale divides the face into nine regions and scores each facial area using a numeric scale (0-5).

Assessments of labiomental crease, marionette lines and the preauricular area from the VISIA images were found to be technically difficult and were therefore excluded from the blinded assessment.

The following six areas were therefore assessed for wrinkles: horizontal forehead lines, glabella frown lines, periorbital lines, cheek folds, nasolabial folds and upper lip lines.

Of the six facial areas which were evaluated, three areas achieved a clinically meaningful result, while three areas did not show improvement. For glabella frown lines, periorbital lines and cheek folds the mean value of change was > 1 grade improvement in the grading scale (see Table v). This change was statistically significant. For horizontal

forehead lines, nasolabial folds, and upper lip lines, the results did not demonstrate a clinically meaningful improvement at day 150.

Table v Comparison of Grade on the Lemperle Scale at Baseline and Day 150 by Three Blinded Assessors (primary endpoint)

	Glabella Frown Lines n (%)	Periorbital Lines n (%)	Cheek Folds n (%)
Subjects graded as having a ≥ 1 grade improvement	41/48 (85.4%)	42/48 (87.5%)	43/48 (89.6%)
Subjects graded as unchanged (Change = 0)	7/48 (14.6%)	6/48 (12.5%)	5/48 (10.4%)
Subjects graded as worsened	0/48 (0%)	0/48 (0%)	0/48 (0%)

For glabella frown lines, periorbital lines and cheek folds an improvement of >1 grade was shown for 18/48 (37.5%), 20/48 (41.7%) and 17/48 (35.4%) subjects (see Table v).

9. Statement of Substantial Equivalence:

MT.DERM GmbH has demonstrated that the Exceed Microneedling device has the same generic classification (generic description) and basic technologies as the predicate device DEN160029. Both devices are micro needling devices containing >1 needle that mechanically punctures or injures the skin for aesthetic use. Both devices have used needle depths of up to 1.5mm in clinical investigations.

Where there are differences between the Exceed micro needling device and the predicate (DEN160029) MT.DERM GmbH has conducted substantial clinical and non-clinical performance testing applicable to those general and special controls deemed necessary by the agency for this product classification and has determined that the Exceed microneedling device raises no new questions relating to safety or effectiveness and therefore has demonstrated that the Exceed microneedling device is substantially equivalent to the referenced predicate DEN160029.